

# Lab Overview

This lab contains four virtual machines: **two Windows machines** and **two Linux machines**. Each platform has a Pivot machine and a Target machine. For practice, use either the Windows Pivot–Target pair or the Linux Pivot–Target pair. Keep the lab network isolated.

## Setup of Windows Pivot-Point Lab

**Pivot IPs:** 10.10.10.30 (Kali) · 10.20.20.20 (VMnet3)

Install and start OpenSSH, set it to start automatically, and allow TCP port 22 through the firewall:

```
Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0
Start-Service sshd
Set-Service sshd -StartupType Automatic
New-NetFirewallRule -Name "OpenSSH-22" -Direction Inbound `
  -Protocol TCP -Action Allow -LocalPort 22
```

Set the RDP and SMB services to start automatically, then start them:

```
Set-Service -Name TermService -StartupType Automatic
Start-Service TermService
Set-Service -Name LanmanServer -StartupType Automatic
Start-Service LanmanServer
```

Enable the RDP and SMB firewall rules:

```
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"
Enable-NetFirewallRule -DisplayGroup "File and Printer Sharing"
```

Create a task to start RDP at boot (run in Command Prompt as Administrator):

```
schtasks /create /tn "WindowsPivot-RDP" ^
  /tr "powershell.exe -Command Start-Service TermService" ^
  /sc onstart /ru SYSTEM /rl HIGHEST /f
```

Create a task to start SMB at boot (run in Command Prompt as Administrator):

Check the Windows Pivot listening ports (SSH, RDP, and SMB):

```
```powershell
Get-NetTCPConnection -State Listen |
  Where-Object { $_.LocalPort -in 22,3389,445 }
```

## Setup of Windows Target

**Target IP:** 10.20.20.110 (VMnet3). The RDP/SMB commands above are for Windows Pivot only.

Install and start OpenSSH, set it to start automatically, and allow TCP port 22 through the firewall:

```
Add-WindowsCapability -Online -Name OpenSSH.Server~~~~0.0.1.0
Start-Service sshd
Set-Service sshd -StartupType Automatic
New-NetFirewallRule -Name "OpenSSH-22" -Direction Inbound `
  -Protocol TCP -Action Allow -LocalPort 22
```

Create the folder for the web-server scripts:

```
New-Item -ItemType Directory -Path C:\Lab -Force
```

Save the Target localhost web-server script as C:\Lab\8080.ps1 :

```

$listener = New-Object System.Net.HttpListener
$listener.Prefixes.Add("http://127.0.0.1:8080/")
$listener.Start()
Write-Host "Windows Target Web Server - 8080 started"
while ($true) {
    $context = $listener.GetContext()
    $html = @"
<!DOCTYPE html>
<html><body>
<h1>Hello from Windows Target</h1>
<p>Port: 8080</p>
</body></html>
"@
    $buffer = [System.Text.Encoding]::UTF8.GetBytes($html)
    $context.Response.ContentType = "text/html; charset=utf-8"
    $context.Response.ContentLength64 = $buffer.Length
    $context.Response.OutputStream.Write($buffer, 0, $buffer.Length)
    $context.Response.OutputStream.Close()
}

```

Save the Target internal web-server script as `C:\Lab\8081.ps1` :

```

$listener = New-Object System.Net.HttpListener
$listener.Prefixes.Add("http://10.20.20.110:8081/")
$listener.Start()
Write-Host "Windows Target Internal Web Server - 8081 started"
while ($true) {
    $context = $listener.GetContext()
    $html = @"
<!DOCTYPE html>
<html><body>
<h1>Hello from Windows Target</h1>
<p>Port: 8081</p>
</body></html>
"@
    $buffer = [System.Text.Encoding]::UTF8.GetBytes($html)
    $context.Response.ContentType = "text/html; charset=utf-8"
    $context.Response.ContentLength64 = $buffer.Length
    $context.Response.OutputStream.Write($buffer, 0, $buffer.Length)
    $context.Response.OutputStream.Close()
}

```

Allow TCP 8081 on the VMnet3 Private firewall profile:

```

New-NetFirewallRule -Name "WindowsTarget-Web-8081" `
    -Direction Inbound -Protocol TCP -Action Allow `
    -LocalPort 8081 -Profile Private

```

Create boot tasks for both Target web scripts (run in Command Prompt as Administrator):

```

schtasks /create /tn "WindowsTarget-8080" ^
    /tr "powershell.exe -ExecutionPolicy Bypass -File C:\Lab\8080.ps1" ^
    /sc onstart /ru SYSTEM /rl HIGHEST /f
schtasks /create /tn "WindowsTarget-8081" ^
    /tr "powershell.exe -ExecutionPolicy Bypass -File C:\Lab\8081.ps1" ^
    /sc onstart /ru SYSTEM /rl HIGHEST /f

```

Start both Target web tasks now:

```
schtasks /run /tn "WindowsTarget-8080"  
schtasks /run /tn "WindowsTarget-8081"
```

Check the listening ports on Windows Target:

```
Get-NetTCPConnection -State Listen |  
Where-Object { $_.LocalPort -in 22,8080,8081 }
```

Test the Target web servers:

```
curl.exe http://127.0.0.1:8080/  
curl.exe http://10.20.20.110:8081/
```

## Setup of Linux Pivot-Point

Start the SSH service:

```
sudo systemctl start ssh
```

Enable SSH to start automatically at boot:

```
sudo systemctl enable ssh
```

Update package lists and install Samba:

```
sudo apt update  
sudo apt install -y samba
```

Start the Samba service:

```
sudo systemctl start smbd
```

Enable Samba to start automatically at boot:

```
sudo systemctl enable smbd
```

Check the SSH and Samba service status:

```
sudo systemctl status ssh smbd --no-pager
```

Check that ports 22 and 445 are listening:

```
sudo ss -ltnp | grep -E ' :22|:445 '
```

Expected ports: **22 = SSH**, **445 = SMB/Samba**.

## Setup of Linux Target

**Linux Target IP:** `10.20.20.100` (VMnet3). The web services bind to all interfaces ( `0.0.0.0` ); keep this VM on the isolated lab network.

Update packages and install OpenSSH Server and Python 3:

```
sudo apt update  
sudo apt install -y openssh-server python3
```

Start SSH and enable it at boot:

```
sudo systemctl start ssh
sudo systemctl enable ssh
```

Create the web root and HTML page for port 8080:

```
sudo mkdir -p /opt/target-8080
sudo tee /opt/target-8080/index.html >/dev/null <<'HTML'
<!DOCTYPE html>
<html><head><title>Linux Target 8080</title></head>
<body><h1>Hello from Linux Target</h1><p>Web Service: Port 8080</p></body>
HTML
```

Create the web root and HTML page for port 8081:

```
sudo mkdir -p /opt/target-8081
sudo tee /opt/target-8081/index.html >/dev/null <<'HTML'
<!DOCTYPE html>
<html><head><title>Linux Target 8081</title></head>
<body><h1>Hello from Linux Target</h1><p>Web Service: Port 8081</p></body>
HTML
```

Create the systemd service file for port 8080:

```
sudo nano /etc/systemd/system/target-8080.service
```

Paste this into the file, then save with **Ctrl+O**, press **Enter**, and exit with **Ctrl+X**:

**[Unit]**

Description=Linux Target Web 8080

After=network.target

**[Service]**

WorkingDirectory=/opt/target-8080

ExecStart=/usr/bin/python3 -m http.server 8080 --bind 0.0.0.0

Restart=always

**[Install]**

WantedBy=multi-user.target

Create the systemd service file for port 8081:

```
sudo nano /etc/systemd/system/target-8081.service
```

Paste this into the file, then save with **Ctrl+O**, press **Enter**, and exit with **Ctrl+X**:

**[Unit]**

Description=Linux Target Web 8081

After=network.target

**[Service]**

WorkingDirectory=/opt/target-8081

ExecStart=/usr/bin/python3 -m http.server 8081 --bind 0.0.0.0

Restart=always

**[Install]**

WantedBy=multi-user.target

Reload systemd, start both web services, and enable them at boot:

```
sudo systemctl daemon-reload
sudo systemctl start target-8080
sudo systemctl start target-8081
sudo systemctl enable target-8080
sudo systemctl enable target-8081
```

Check SSH and both web-service statuses:

```
sudo systemctl status ssh target-8080 target-8081 --no-pager
```

Check that ports 22, 8080, and 8081 are listening:

```
sudo ss -ltnp | grep -E ' :22|:8080|:8081 '
```

Expected: **22 = SSH, 8080 = Web, 8081 = Web.**

Test the pages in a browser:

- Port 8080: `http://10.20.20.100:8080`
- Port 8081: `http://10.20.20.100:8081`

After reboot, SSH and both web services will start automatically.